CBCS SCHEME

22SDS21

USN | | | | | | | | | | |

Second Semester M.Tech Degree Examination, June/July 2024

Deep Learning

Time: 3 hrs. Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.

			M	L	C
		Module – 1			
Q.1	a.	Explain the NO free launch theorem.	10	L2	CO1
	b.	Define Deep Learning? Differentiate machine learning and deep learning with example.	10	L2	CO2
		OR			
Q.2	a.	Explain briefly about unsupervised learning approach like PCA and K-means clustering with suitable example.	10	L2	CO2
	b.	Explain maximum likelihood estimation. Define condition log likelihood, MSE.	10	L2	CO1
		Module – 2			
Q.3	a.	Explain the concept of 'Ensemble learning' with an example.	8	L2	CO2
	b.	What are deep feed forward networks? Explain the gradient based learning.	8	L2	CO1
	c.	What is Adversarial training?	4	L1	CO1
		OR			
Q.4	a.	Explain briefly about hidden units.	10	L2	CO1
	b.	Explain briefly about Dropout.	10	L2	CO1
		Module – 3			
Q.5	a.	Explain the challenges faced while optimizing a neural network modes.	10	L2	CO1
	b.	Define Max pooling. Explain the three stages of convolution network with a neat diagram.	10	L2	CO1
		OR			
Q.6	a.	Explain how learning differs from pure optimization explain any two.	10	L2	CO2
	b.	Illustrate convolution operation.	10	L3	CO1
		Module – 4			
Q.7	a.	Explain Recurrent Neural Network.	10	L2	CO1
	b.	Explain LSTM algorithm with suitable diagram.	10	L2	CO1
		OR			
Q.8	a.	Illustrate how computation is done in a typical bidirectional graph.	10	L2	CO1
	b.	Explain the encoder – decoder sequence to sequence architecture.	10	L2	CO1
		Module – 5			
Q.9	a.	Name the give ways of selecting hyper parameters. Explain any one in detail.	10	L1	CO2
	b.	What is preprocessing. Explain contract normalization in case of computer vision.	10	L1	CO1
		OR			
Q.10	a.	What is natural language processing? Explain n-grams with an example.	10	L2	CO2
	b.	Define the following performance metric : i) Accuracy ii) Precision iii) Recall iv) F-Score v) Coverage.	10	L1	CO1

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